

Nuclei Segmentation Reimagined.

Hybrid CNN-Transformer Architectures for Digital Pathology

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CONFERENCE PAPER

Why is this hard?

Nuclei segmentation in histopathology faces four compounding challenges.

01

Dense Overlap

Closely packed leukaemia cells defeat classical watershed algorithms.

02

Morphological Diversity

Size, shape, chromatin texture vary dramatically across cell types.

03

Staining Variability

H&E protocols differ lab-to-lab, causing domain shift.

04

Tissue Heterogeneity

Breast tissue models fail on lung or colon – no free generalization.

Cancer grading depends on every cell.

190K+

labeled nuclei · PanNuke

19

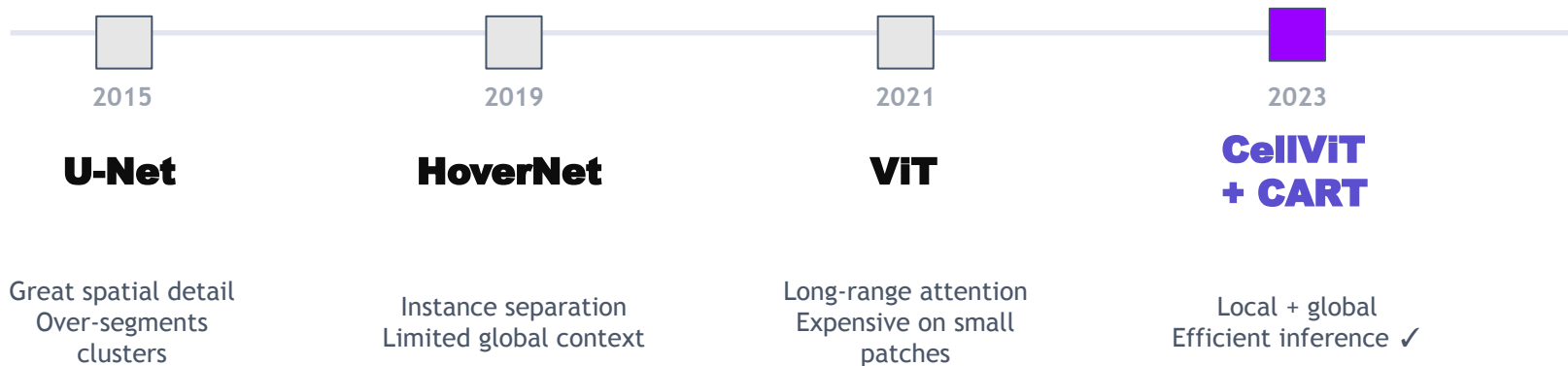
tissue types covered

5

nucleus classes segmented

Accurate nuclei maps unlock tumor microenvironment analysis, prognosis modeling, and precision oncology.

The field needed a bridge.



One encoder. Three

ViT-Small Encoder

ImageNet pre-trained · Patch embeddings
Transformer blocks → feature maps

CART replaces last-half MHSA

Classification

5 nucleus
classes

HV Maps

Instance
separation

Binary Mask

Nucleus
detection

Multi-task Loss

$$L_{\text{total}} = \lambda_1 \cdot L_{\text{cls}} + \lambda_2 \cdot L_{\text{bin}} + \lambda_3 \cdot L_{\text{hv}}$$

L_{cls}

Class-weighted cross-entropy · penalizes
minority classes harder

L_{bin}

Binary segmentation · nucleus vs. background

L_{hv}

Horizontal + vertical distance regression ·
enables instance splits

CART: Transformers that think like CNNs.

Context-Aware Re-Parameterizable Transformer – our novel contribution.

01

Training

Two branches run in parallel:
3×3 kernels (local texture) +
1×1 kernels (channel mixing).
Learns multi-scale features.

02

Re-param

At convergence, 1×1 weights
are zero-padded to 3×3 and
added to the 3×3 branch.
 $y = (W_{3\times3} + \text{Pad}(W_{1\times1})) * x$

03

Inference

Single 3×3 convolution.
Zero overhead vs standard
CNN – deployment-ready
for clinical edge devices.

What changes when we unfreeze?

Constrained

- ✗ Frozen ViT backbone
- ✗ Decoder branches only
- ✗ 10 epochs
- ✗ Standard cross-entropy
- ✗ ImageNet features only

vs

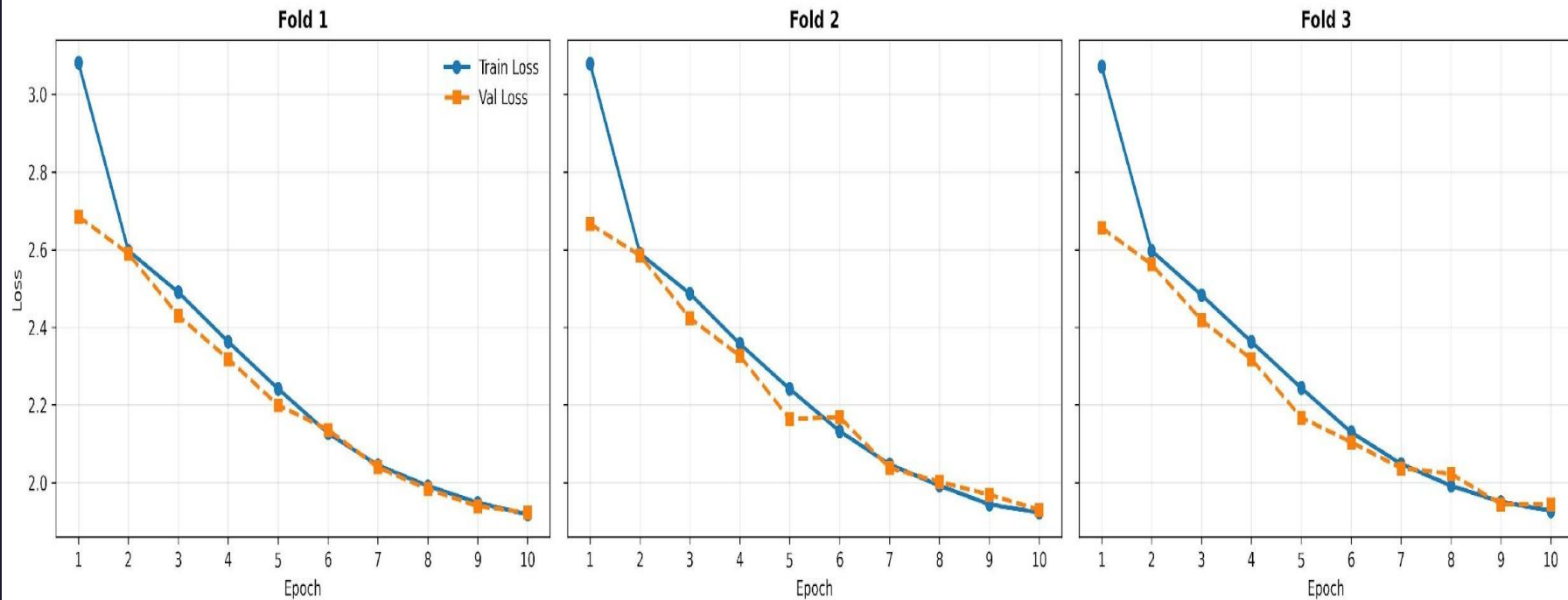
Improved

- ✓ Unfrozen ViT backbone
- ✓ Full end-to-end training
- ✓ 30 epochs + early stopping
- ✓ Class-weighted loss
- ✓ AMP for training stability

Training Curves

CellViT on PanNuke · 3-Fold Cross-Validation · 10 Epochs each

CellViT Reproduction on PanNuke: Training Curves



Loss converges consistently across all 3 folds – validation tracks training closely, no overfitting

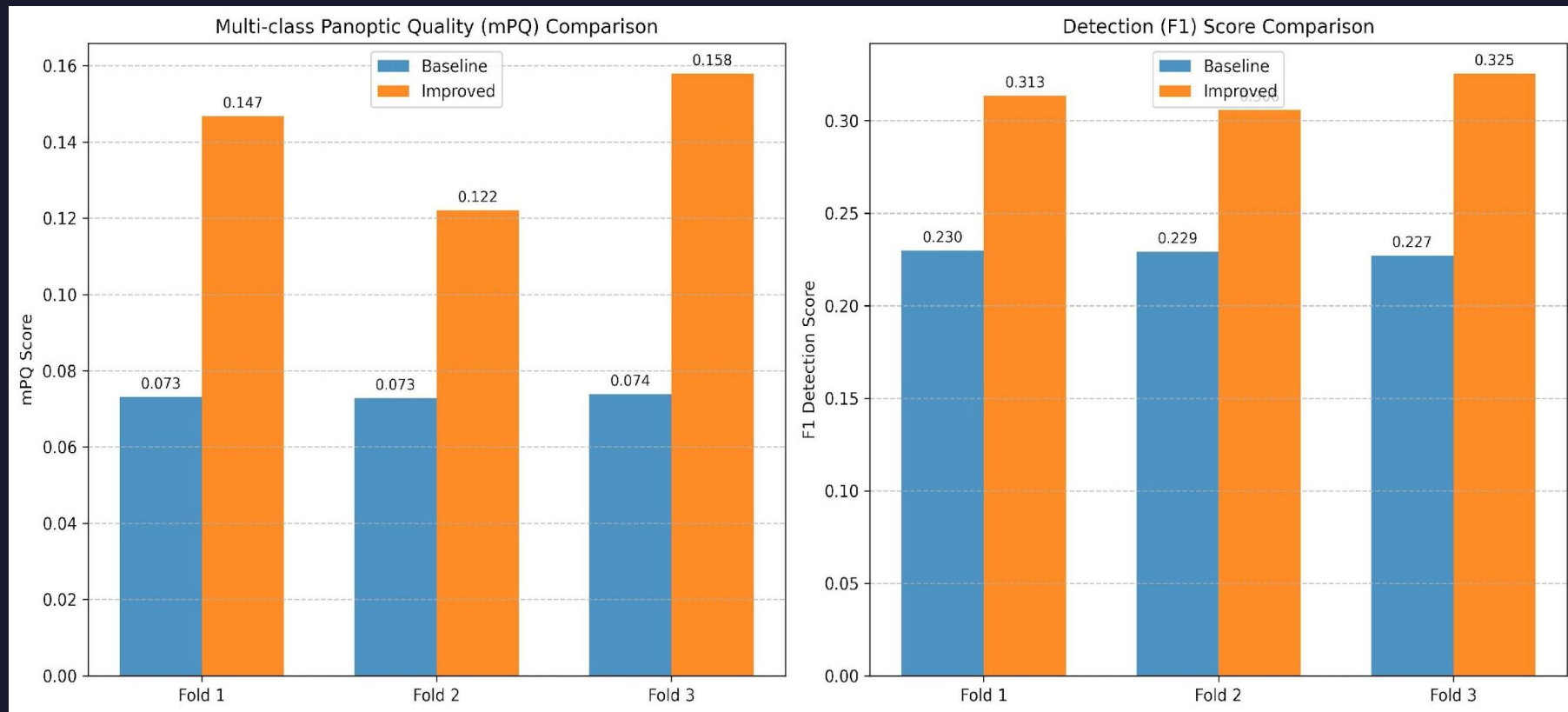
+94%

improvement in Mean Panoptic Quality

Frozen baseline: 7.32% → Improved pipeline: 14.22%

Fold-by-Fold Performance Gains

Baseline (frozen ViT) vs Improved (unfrozen + class-weighted loss) · mPQ and F1 Score



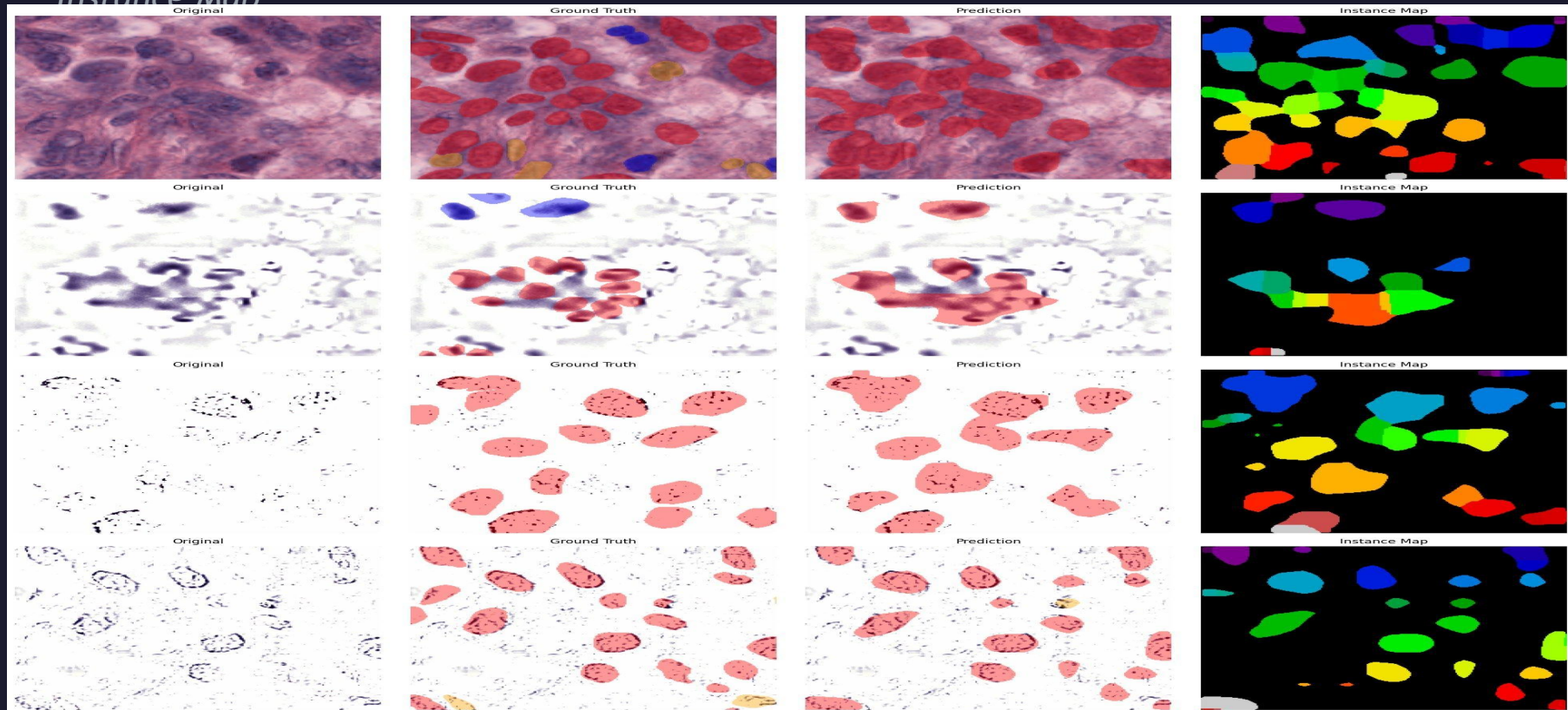
Improved model doubles mPQ (0.073→0.14+) and boosts F1 detection (0.23→0.32+) across all folds

Consistent gains across all three folds.

Fold	Constrained mPQ	Improved mPQ	Gain	Paper Target
Fold 1	7.30%	14.67%	+101%	47.88%
Fold 2	7.28%	12.20%	+68%	47.76%
Fold 3	7.38%	15.79%	+114%	49.15%
Mean	7.32%	14.22%	+94%	48.27%

Sample Predictions: Baseline

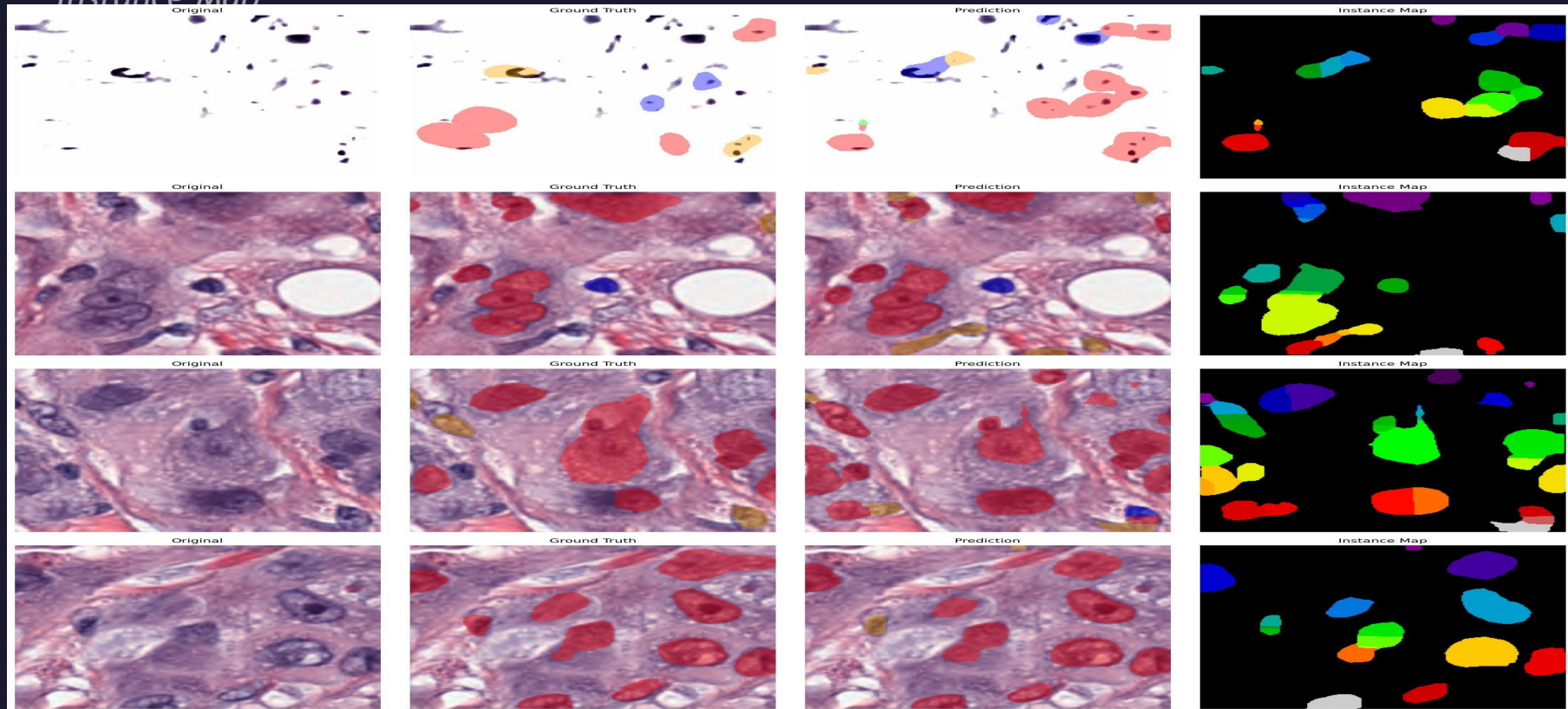
Frozen ViT backbone · Standard cross-entropy loss · Original | Ground Truth | Prediction | Instance Map



Baseline struggles with class boundaries and instance separation — epithelial and connective tissue often merged

Sample Predictions: Improved

Unfrozen ViT backbone · Class-weighted loss · Original | Ground Truth | Prediction | Instance Map



Improved model captures finer boundaries, better nuclei class discrimination and cleaner instance separation

Key Conclusions

01

Hybrid architectures are essential for scalable digital pathology.

02

Unfreeze the backbone – ImageNet features alone can't capture chromatin.

03

Class-weighted loss rescues minority classes (Dead +2.5×).

04

Strong generalization across unseen staining and tissue types.

05

CART proves Transformer quality at CNN inference cost.

Thank You.

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